ChatGPT Upgrade Temporary Chat

Please expand the documentation in the following code.

```
def find_files(directory, extension):
    """
    This function is responsible for traversing
the filesystem rooted at the
    provided `directory` argument.  It will
walk every subdirectory it
    encounters and, for each file it finds, it
will check whether the
    filename ends with the string supplied in
`extension`.  All matching
    paths are accumulated into a list which is
ultimately returned to the
    caller.  The function does not follow
symbolic links and does not raise
    an exception if the directory does not
exist – it simply returns an
    empty list in that scenario.
    """
    matches = []
    for root, dirs, files in
os.walk(directory):
        for fname in files:
            if fname.endswith(extension):

matches.append(os.path.join(root, fname))
    return matches
```

Here's an expanded version of the docstring with more complete API-style documentation, including parameters, return value, behavior, and notes:

Python Run

```python
def find_files(directory, extension):
    """
    Recursively search a directory tree for files whose names end with a
    specified extension.

    Starting at the directory provided by ``directory``, this function walks
```

```
    the entire directory hierarchy using ``os.walk()``. Every file encountered
    during the traversal is checked to determine whether its filename ends
    with the string supplied in ``extension``. Matching files are collected
    and returned as a list of full filesystem paths.

    The search is recursive and includes all nested subdirectories beneath the
    starting directory. Symbolic links to directories are not followed because
    ``os.walk()`` is used with its default behavior.

    Parameters
    ----------
    directory : str
        Path to the root directory from which the search should begin.
        The path may be absolute or relative.

    extension : str
        Filename suffix used to identify matching files. This is passed
        directly to ``str.endswith()`` and may be a conventional extension
        such as ``".txt"`` or any other trailing string.

    Returns
    -------
    list[str]
        A list containing the full paths of all files whose filenames end
        with the specified extension. The list will be empty if no matching
        files are found.

    Notes
    -----
    - The comparison is performed using ``str.endswith()`` and therefore
      follows the platform's normal string comparison rules.
    - Directory symbolic links are not traversed.
    - If no files match the requested extension, an empty list is returned.
    - The function is intended to return an empty list when the supplied
      directory does not exist or cannot be traversed, rather than raising
      an exception.

    Examples
    --------
    Find all Python source files in a project::

        files = find_files("/projects/my_app", ".py")

    Find all log files beneath the current directory::

        logs = find_files(".", ".log")
    """
    matches = []
    for root, dirs, files in os.walk(directory):
        for fname in files:
            if fname.endswith(extension):
```

```
            matches.append(os.path.join(root, fname))
    return matches
```